

PYTHON PROJECT

PERSONAL EXPENSE TRACKER

TRACE

MANAGE

ANALYSE

SAVE

A simple and simple console based financial management system

Submitted by: Amisha Patel

Registration no : 26BCE10105

Course: CSE CORE

Semester: First

Teacher: Adarsh Patel

Github Repository Link: [amisha26bce/Expense-Tracker](https://github.com/amisha26bce/Expense-Tracker)

Declaration

I hereby declare that this project report titled Personal Expense Tracker has been prepared as part of the Python programming project work. The application demonstrates functions, modules, conditional statements, loops, lists, dictionaries, exception handling and JSON file handling.

ABSTRACT

Personal Expense Tracker is a Python-based console application designed to help a user record, manage and analyse personal expenses. The system allows profile creation, expense entry, viewing, editing and deletion of records, monthly budget management, spending analysis and searching/filtering. The application uses JSON file handling for persistent storage, which means that expense records remain available after the program is closed. The project is divided into separate modules so that the code is organised, readable and easier to maintain.

OBJECTIVES

1 Develop a simple and user-friendly expense management application.

2 Record daily expenses with amount, category, description and date.

3 View, edit and delete expense records.

4 Set a monthly budget and monitor spending.

5 Calculate total spending and identify the highest expense.

6 Generate category-wise spending information.

7 Search expenses by category, date or description.

8 Store data permanently using JSON.

9 Demonstrate modular Python programming.

MAJOR FEATURES

Add Expense Record amount, category, description and date

View Expenses Display all saved expense records.

Edit Expense Update an existing expense using its ID.

Delete Expense Remove an unwanted expense record.

Monthly Budget Set and monitor a monthly spending limit.

Expense Analysis Calculate total, count, highest expense and category totals.

Search / Filter Find expenses by category, date or description.

Persistent Storage Save profile, expenses and budget in JSON.

SYSTEM MODULES

`main.py`

Entry Point

Starts the application, displays the welcome message, creates the profile and opens the main menu.

`data.py`

Data Management

Handles JSON storage, profile creation, adding/viewing/editing/deleting expenses and setting the budget.

`menus.py`

User Interface

Displays the main menu and connects each user choice to the required function.

`analysis.py`

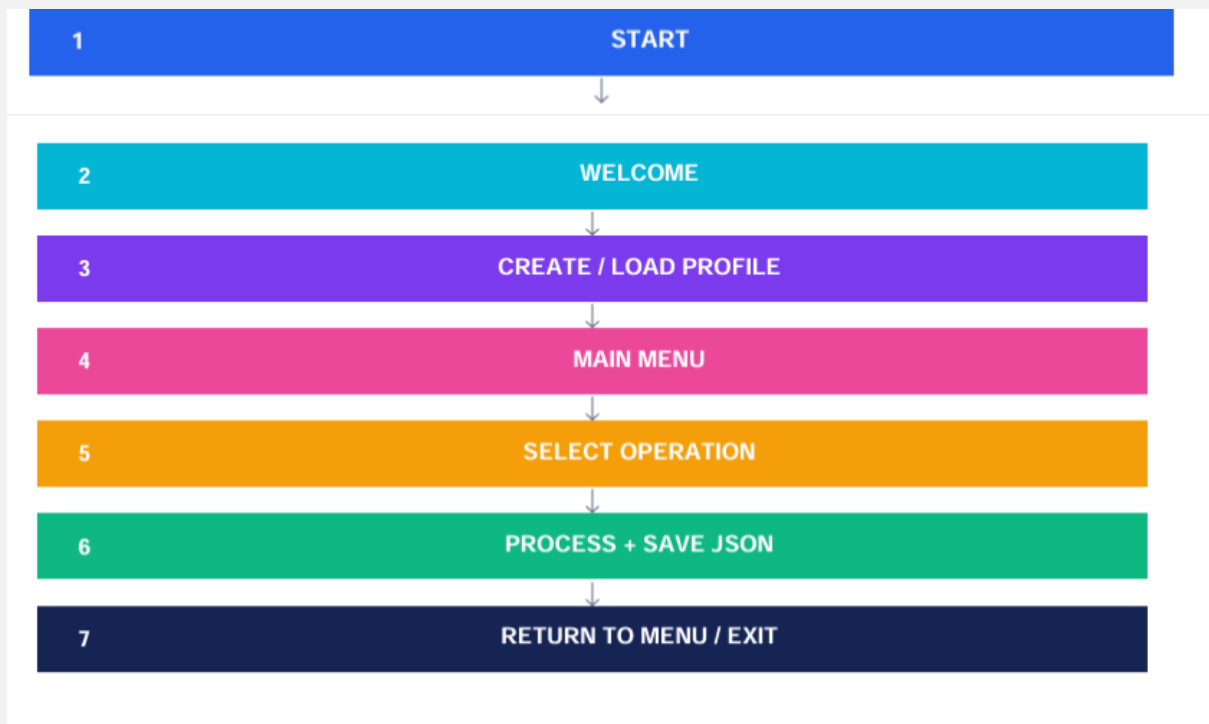
Analysis & Search

Calculates totals, budget usage, highest expense, category totals and provides search/filter operations.

TECHNOLOGY AND CONCEPT

Concept	Purpose
Python	Core programming language
Functions	Reusable operations and modular code
Lists & Dictionaries	Store multiple records and structured data
JSON	Permanent storage of project data
File Handling	Read/write expenses.json
Exception Handling	Handle invalid input and file errors
datetime	Use current date for expenses
Modules	Separate data, menus and analysis

PROGRAM FLOW



DATA STORAGE

The application stores its information in expenses.json. The file contains the user's profile, the list of expenses and the monthly budget.

TESTING

Test	Expected Result	Status
Create profile	Name and ID are saved	PASS
Add valid expense	Record is stored	PASS
Invalid amount	Error message appears	PASS
View expenses	Saved records display	PASS
Edit expense	Selected record updates	PASS
Delete expense	Selected record is removed	PASS
Set budget	Budget is saved	PASS
Analysis	Totals and category data show	PASS
Search	Matching records display	PASS
Exit	Program closes normally	PASS

ADVANTAGES

- 1) Beginner-friendly console interface.
- 2) Permanent storage through JSON.
- 3) Modular and maintainable code.
- 4) Supports basic budget monitoring.
- 5) Includes validation for important numeric input.

FUTURE SCOPE

Add a graphical user interface (GUI).

Add charts and graphs for visual analysis

Generate monthly and yearly reports

Export reports to CSV/PDF.

Authentication and multiple users.

Add recurring expense reminders.

CONCLUSION

The Personal Expense Tracker demonstrates how Python can be used to create a practical application using modular programming and file handling. It combines expense CRUD operations, budget tracking, searching and basic analysis in a single project. The structure also provides a clear base for future enhancements such as graphical dashboards and advanced reporting.